

# OS BANANA-4 ASSIGNMENT

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SEC: F

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1A

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

1B

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ |
```

2A

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization
Stored tree object hash: 26198bf54d24ffda25c593cf008d4e99171cbe647eff872540c3b8f4dd3a0446
Object stored at: .pes/objects/26/198bf54d24ffda25c593cf008d4e99171cbe647eff872540c3b8f4dd3a0446
PASS: tree object store roundtrip

All Phase 2 tests passed.
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ |
```

2B

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ xxd
.pes/objects/26/198bf54d24ffda25c593cf008d4e99171cbe647eff872540c3b8f4dd3a0446
00000000: 7472 6565 2039 3000 3130 3036 3434 2061  tree 90.100644 a
00000010: 2e74 7874 00dd dddd dddd dddd dddd dddd  .txt.....
00000020: dddd dddd dddd dddd dddd dddd dddd dddd  .....
00000030: dddd dddd dd31 3030 3634 3420 622e 7478  ....100644 b.tx
00000040: 7400 eeee eeee eeee eeee eeee eeee eeee  t.....
00000050: eeee eeee eeee eeee eeee eeee eeee eeee  .....
00000060: eeee ..
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$
```

3A

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Staged changes:
  staged:    file1.txt
  staged:    file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked: commit.h
  untracked: test_objects.c
  untracked: commit.c
  untracked: pes.h
  untracked: object.c
  untracked: index.h
  untracked: .gitignore
  untracked: pes.c
  untracked: Makefile
  untracked: README.md
  untracked: tree.c
  untracked: test_tree.c
  untracked: test_tree
  untracked: test_sequence.sh
  untracked: tree.h
  untracked: test_objects
  untracked: index.c
```

3B

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776669963 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776669963 6 file2.txt
```

4A

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ ./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: df0c53d69195... Initial commit
Committed: 350ef98ae23a... Add world
Committed: e1f3ef969fcd... Add farewell
commit e1f3ef969fcd67e6bb35a19ec2fa24b20a6d4eab3739c8fc0a1bd0c13531cbf
Author: Piyush <PES2UG24CS350>
Date: 1776670122

    Add farewell

commit 350ef98ae23acbd28fb1c85a568a50674676e4468aeb79f780ed6072512fa224
Author: Piyush <PES2UG24CS350>
Date: 1776670122

    Add world

commit df0c53d69195bda2aea26835a07e0beb1f68f62fdb65300e026b129d0bbb5bdf
Author: Piyush <PES2UG24CS350>
Date: 1776670122

    Initial commit
```

4B

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/26/198bf54d24ffda25c593cf008d4e99171cbe647eff872540c3b8f4dd3a0446
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
.pes/objects/35/0ef98ae23acbd28fb1c85a568a50674676e4468aeb79f780ed6072512fa224
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/85/e367b76c2417b47a95f46208835ad8700dd58335171c94a497974be6191900
.pes/objects/96/c0d5c28dd3b719606c36723cd16aefbc73b3dc8b74de61070183357d6968b9
.pes/objects/df/0c53d69195bda2aea26835a07e0beb1f68f62fdb65300e026b129d0bbb5bdf
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e1/f3ef969fcd67e6bb35a19ec2fa24b20a6d4eab3739c8fc0a1bd0c13531cbf
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f1/0239b620372163c2dd06ac33189180c55edf1c56b15990a1cf44e00b1b0e59
.pes/refs/heads/main
```

4C

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ cat .pes/refs/heads/main
e1f3ef969fcdf67ebbc35a19ec2fa24b20a6d4eab3739c8fc0a1bd0c13531cbf
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

## Final: (Integration Test)

```
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ make test-integration
Makefile:47: *** missing separator. Stop.
piyush-a-patel@piyush:~/Patel/Assignment/OS/Orange-4/PES2UG24CS350-pes-vcs$ bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:    file.txt
  staged:    hello.txt

Unstaged changes:
(nothing to show)

Untracked files:
(nothing to show)

--- First Commit ---
Committed: 7e7d2307796d... Initial commit

Log after first commit:
commit 7e7d2307796dc820872b4c1f4a6bc9722ad1fb5808edfaea6ea45584a6855fb9
Author: PES User <pes@localhost>
Date:   1776677455

    Initial commit

--- Second Commit ---
Committed: d1b506f81f00... Update file.txt

--- Third Commit ---
Committed: 3f93b19ea8bd... Add farewell

--- Full History ---
commit 3f93b19ea8bdf1df7ee61c55692902b3d84b6be8d587e27d88f29ad877e491e8
Author: PES User <pes@localhost>
Date:   1776677455

    Add farewell

commit d1b506f81f0087ed5ef13301feac32861c0a2e4ee616a72a50b085ba5e414f5e
Author: PES User <pes@localhost>
Date:   1776677455

    Update file.txt

commit 7e7d2307796dc820872b4c1f4a6bc9722ad1fb5808edfaea6ea45584a6855fb9
Author: PES User <pes@localhost>
Date:   1776677455

    Initial commit
```

```
--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
3f93b19ea8bdf1df7ee61c55692902b3d84b6be8d587e27d88f29ad877e491e8

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/3f/93b19ea8bdf1df7ee61c55692902b3d84b6be8d587e27d88f29ad877e491e8
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/7e/7d2307796dc820872b4c1f4a6bc9722ad1fb5808edfaea6ea45584a6855fb9
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf9597f88299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafaa3829f61f75b8d10d5350f9
.pes/objects/d1/b506f81f0087ed5ef13301feac32861c0a2e4ee616a72a50b085ba5e414f5e
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
test_sequence.sh: line 82: //: Is a directory
```

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### Q5.1

A checkout operation updates both repository metadata and the working directory. In .pes, the HEAD file must be updated to point to the selected branch, typically in the form "ref: refs/heads/branchname". The branch file itself already contains the commit hash, so it does not change.

The system then resolves the branch to a commit, reads the commit object to obtain its tree, and reconstructs the working directory from that tree. This involves creating files that exist in the target tree, overwriting files that differ, and deleting files that are not present in the target tree. After updating the working directory, the index must also be rewritten so that it reflects the new tree state.

The operation is complex because it must avoid overwriting uncommitted changes. It requires comparing the working directory, index, and target tree. It also has to handle directory structures, file mode differences, and ensure the update is consistent even if interrupted.

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### Q5.2

To detect a dirty working directory, compare each tracked file in the index with the current working directory version. For each entry, read the file from disk, compute its hash using the same object hashing method, and compare it with the hash stored in the index. If they differ, the file has been modified.

Next, compare the index with the target branch's tree. If a file differs between the current index and the target tree, and also differs between the index and the working directory, then there is a conflict. In this case, checkout must be refused to prevent loss of uncommitted changes.

This approach uses only the index and object store, without needing extra metadata.

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### Q5.3

In a detached HEAD state, HEAD directly stores a commit hash instead of pointing to a branch. If new commits are created in this state, they are not

referenced by any branch. They still exist in the object store, but no named reference points to them.

These commits can be recovered if the user still knows their hash or if they are visible through mechanisms like a reflog. The user can create a new branch pointing to that commit, making it reachable again. If no reference is created, the commits may eventually be removed by garbage collection.

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#### Q6.1

To find unreachable objects, start from all branch references in refs/heads and mark their commit hashes as reachable. Traverse each commit by following its parent links and associated tree objects. For each tree, recursively visit all subtree objects and blobs. Mark every visited object as reachable.

After traversal, iterate through all objects in the object store and delete those not marked as reachable.

A hash set is an efficient data structure for tracking reachable objects, as it allows constant-time insertion and lookup.

For a repository with 100000 commits and 50 branches, the number of unique objects visited is approximately the number of commits plus associated trees and blobs. Since many commits share history, the traversal will likely visit around 100000 commits, not 50 times that number. Including trees and blobs, the total could be a few hundred thousand objects.

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#### Q6.2

Running garbage collection concurrently with a commit operation is dangerous because of race conditions. For example, a commit operation may create new objects (blobs, trees) and compute their hashes, but before the commit reference is written, these objects are not yet reachable from any branch. If garbage collection runs at that moment, it may consider these objects unreachable and delete them.

This leads to a corrupted commit that references missing objects.

Git avoids this by ensuring that new objects are safely written and referenced before they can be collected. It uses mechanisms such as temporary files, atomic renaming, and object protection during operations. Additionally, Git's garbage collection is typically not run concurrently with active writes, or it uses safeguards like packfiles and reachability checks that prevent recently created objects from being deleted prematurely.

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